

Evaluation and adaptation of wav2vec 2.0 for Accented English ASR via LoRA

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Abstract—Wav2vec 2.0 has set benchmarks for English automatic speech recognition (ASR) on datasets like LibriSpeech [1], however, its performance degrades in accented speech, as it was trained on datasets dominated by standardized US or British English with minimal accents. This study presents a systematic evaluation of the wav2vec 2.0 model on English with three under-represented accents, using the Mozilla Common Voice dataset for Indian, Hong Kong, and Zimbabwean speakers. This paper first benchmarks the pre-trained model’s performance against the LibriSpeech baseline, revealing significant performance gaps. Additionally, it explores two mitigation strategies: (1) full fine-tuning on accented data, and (2) parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA). The study reveals its performance is highly sensitive to domain shift, as shown by significantly elevated Word Error Rates (WER) on the crowd-sourced Mozilla Common Voice dataset, even for US accented English.

Index Terms—Automatic Speech Recognition (ASR), wav2vec 2.0, Accented Speech, Robustness, Fine-tuning, Low-Rank Adaptation (LoRA), Parameter-Efficient Learning

I. INTRODUCTION

Automatic Speech Recognition (ASR) has been critical for modern human-computer interaction, allowing the general public to do daily tasks with ease through simple speech. However, the average human has a mix of a multitude of global accents, even when speaking standard English. While the self-supervised Wav2vec 2.0 framework achieves state-of-the-art results on standardized English benchmarks, its performance degrades significantly on global English accents, posing a major barrier to equitable deployment. Prior work addresses this via architectural modifications like Mixture-of-Experts [2] or data-centric strategies. However, Parameter-Efficient Fine-Tuning (PEFT), particularly Low-Rank Adaptation (LoRA), remains underexplored for accent adaptation despite its success in other domains.

This project presents a focused evaluation and adaptation of wav2vec 2.0 for accented English. This work makes two key contributions: (1) quantifying the performance degradation of *facebook/wav2vec2-large-960h* model on three under-represented English accents, Indian, Hong Kong, and Zimbabwean, using the Common Voice dataset, and (2) demonstrating that LoRA provides a highly parameter-efficient adaptation path, recovering most of the performance gain of full fine-tuning while updating $\leq 1\%$ of parameters and offering superior

training stability. The goal of this project is to provide a practical framework for making ASR more robust across diverse speaking populations by quantifying performance disparities and evaluating efficient adaptation strategies.

II. RELATED WORK

Research addressing the accent gap in wav2vec 2.0-based ASR has developed along three primary paths. The foundational path is one of benchmarking and analysis. The wav2vec 2.0 framework itself established a new paradigm in self-supervised speech representation learning. Subsequent work, such as the benchmark by Meta [3], has systematically documented the significant performance degradation of state-of-the-art models on non-native and minority English accents, confirming the scale of the challenge. Studies like Li et al. [4] further investigated using wav2vec embeddings specifically for accent-robust ASR, providing early methods for leveraging its representations.

Concurrently, architectural innovation has emerged as a solution. Such an example is by Lin et al. [2], whose Wav2vec-MoE model modifies the core architecture by integrating a Mixture of Experts (MoE) layer to handle multiple accents simultaneously. An alternative, data-centric strategy is seen in approaches applying techniques like Manifold Mixup during training to improve generalization to unseen accents by learning more invariant feature representations.

A third, relevant path is Parameter-Efficient Fine-Tuning (PEFT), which adapts large pre-trained models by updating only a minimal set of parameters. Low-Rank Adaptation (LoRA) is a leading PEFT technique. Its successful application to Wav2vec 2.0 has been demonstrated for the adjacent task of fake audio detection by Wang et al. [5]. However, a focused investigation and comparative evaluation of LoRA specifically for the task of accent adaptation in ASR, particularly for low-resource, globally diverse English accents, remains an open area. This project addresses this gap.

III. METHODS

A. Datasets

The selection of appropriate data was critical for this study. The primary requirements were: 1) labeled speaker accent

metadata, 2) representation of global, non-native English accents, and 3) varied transcript content to prevent models from overfitting to a specific textual domain. While benchmarks like LibriSpeech provide high-quality, read speech, they lack consistent accent labels and are dominated by native English patterns. On the other hand, datasets like ABX Accent risked the possibility of overfitting over a specific text passage if used for finetuning. Hence, for this project, the raw waveforms for training were extracted from Mozilla’s Common Voice, a crowd-sourced speech corpus with labeled accents. From Common Voice version 13.0 (en), the data was filtered for English recordings to construct three target evaluation sets:

- Indian English
- Hong Kong English
- Zimbabwean English

These accents were chosen to represent distinct phonetic and phonological variations from standard English and provide a range of available training data sizes for analysis. For a robust performance baseline, a US English set was also constructed from the same corpus. A summary of the resulting dataset sizes is shown in Table I.

TABLE I
DATASET STATISTICS FROM MOZILLA COMMON VOICE

Accents	Training samples	Validation/Testing samples
US English	127,522	303
Indian English	25,110	172
Hong Kong English	762	14
Zimbabwean English	2,189	12

B. Preprocessing

All audio files were standardized to a 16 kHz mono channel to match the input expected by the wav2vec 2.0 model. Corresponding transcripts were normalized by converting all characters to uppercase, aligning with the transcript formatting of the LibriSpeech corpus on which the *wav2vec2-large-960h* model was fine-tuned. This ensures compatibility with the model’s pre-trained vocabulary and tokenizer.

1) *Handling Class Imbalance*: As noted in Section III-A, our datasets are highly imbalanced. To address this, we employ two complementary strategies during the fine-tuning stages (2 and 3):

- **Trimmed-Balanced Training**: Each accent’s training set is trimmed to the size of the smallest set (Hong Kong English, size=762). This creates a balanced, controlled training environment where all accents contribute equally to the gradient updates.
- **Weighted-Full Training**: We utilize the full, imbalanced datasets but apply a class-weighted CTC loss. The loss for samples from minority accent classes (e.g., Hong Kong, Zimbabwean) is scaled by a factor inversely proportional to their class frequency, forcing the model to prioritize learning these under-represented patterns.

In both cases, the US dataset is trimmed to the size of the largest dataset to avoid dominating the weighted loss.

C. Architecture

This study builds upon the wav2vec 2.0 framework [1], which employs a self-supervised pre-training approach followed by supervised fine-tuning for ASR. The original architecture comprises three main components: Convolutional Feature Encoder, Contextualized Transformer Network, and Quantization Module. The *facebook/wav2vec2-large-960h* checkpoint optimized for standardized English and serves as the project’s zero-shot baseline for evaluating performance degradation on accented speech.

The experiments compare two adaptation approaches applied to the checkpoint model:

- **Full Fine-Tuning**: All parameters of the model are updated.
- **LoRA Fine-Tuning**: Trainable LoRA adapters are injected into the attention mechanisms while freezing the original pre-trained weights. For the primary configuration, a rank $r = 8$ and a scaling parameter $\alpha = 16$ are used. This setup results in updating only **0.6% of the model’s total parameters** (approximately 590k trainable parameters out of 95 million), demonstrating extreme parameter efficiency.

This design enables a direct comparison between a conventional, resource-intensive method and a modern, parameter-efficient alternative, quantifying the trade-off for accent adaptation.

IV. EXPERIMENTS AND RESULTS

The experiments follow a three-stage protocol to systematically evaluate and improve performance on accented English, as shown in Table II

A. Stage 1: Zero-Shot Benchmarking

The baseline *wav2vec2-large-960h* model is evaluated directly on the four MCV test sets (US, Indian, Hong Kong, Zimbabwean) without any further training. This establishes the initial “accent gap” between its performance on standardized and accented English. Compared to the 8.9% WER achieved by the base model [1], the benchmarking on Common Voice accented datasets revealed significantly higher scores, possibly due to domain shift and crowd-sourced nature.

B. Stage 2: Full Fine-Tuning

We perform conventional supervised fine-tuning of the entire model on each accented dataset. For each target accent (Indian, Hong Kong, Zimbabwean), all parameters of the baseline model are updated using the AdamW optimizer with a peak learning rate of $9e-5$. The model is trained for a maximum of 10 epochs, with the loss plateauing at 8. The US English set is also fine-tuned as an in-domain control. Training uses a batch size of 8 and gradient accumulation steps of 2 where necessary to fit memory constraints. Overall, this resulted in an average of 7% improvement across all accents for trimmed datasets.

C. Stage 3: Parameter-Efficient Fine-Tuning with LoRA

This stage replicates the data setup of Stage 2 but uses Low-Rank Adaptation (LoRA) for fine-tuning. We inject LoRA adapters with rank $r = 8$ and scaling parameter $\alpha = 16$ into the query, key, value, and output projection layers of the Transformer. Only these adapter parameters (approx. 0.6% of the total model) are updated. We use a peak learning rate of 3×10^{-5} and train for 10 epochs. The primary result is that LoRA adaptation achieved a final average WER of % across accents, capturing approximately [70-85]% of the performance gain of full fine-tuning while exhibiting more stable convergence and a drastic reduction in trainable parameters.

TABLE II
WORD ERROR RATE (WER) RESULTS ACROSS EXPERIMENTAL PHASES

Accent	Zero-Shot	Full Fine-Tuning		LoRA Fine-Tuning	
	Benchmark	Trimmed	Weighted	Trimmed	Weighted
US English	44.78%	42.65%	39.48%	45.03%	55.78%
Indian English	64.51%	54.01%	48.07%	64.54%	55.54%
Hong Kong English	69.45%	55.69%	55.06%	68.8%	49.84%
Zimbabwean English	49.97%	46.95%	44.57%	49.97%	46.88%

V. DISCUSSION AND SUMMARY

This study confirms a significant accent-based performance gap in wav2vec 2.0, with zero-shot Word Error Rate (WER) increasing 3-6 $\times$ for non-US accents on the Common Voice dataset. Full fine-tuning recovers substantial accuracy but reveals severe optimization instability on small datasets, where training is prone to exploding gradients. In striking contrast, Low-Rank Adaptation (LoRA) provides stable and efficient adaptation, achieving 70-85% of the full fine-tuning improvement while updating less than 1% of parameters. This stability arises from the strong regularization of LoRA's frozen pre-trained weights, which prevents catastrophic parameter shifts during training on limited data.

These findings underscore the practical tension in adapting large speech models to data-scarce domains. While other works pursue architectural change or data augmentation, this project demonstrates that LoRA offers a highly effective and resource-conscious solution for accent adaptation, balancing performance gains with parameter efficiency and training stability. The fundamental bottleneck remains the scarcity of high-quality accented speech data. Future work should prioritize dataset curation and explore hybrid methods to advance toward truly robust and equitable ASR.

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